MILITARY SPECIFICATION

RADIO SET AN/URC-94(V)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design and performance of the radio set AN/URC-94(V), a multimode high frequency (HF) and low band very high frequency (VHF) communication transceiver operating in the 1.5 to 29.9999 megahertz (MHz) frequency band in the amplitude modulation (AM) mode and in the 30.0 to 79.9999 MHz frequency band in the frequency modulation (FM) mode. The AN/URC-94(V) is a militarized version of a commercial equipment, the radio frequency (RF) communications model RF-280.

1.2 This specification covers two variations of the AN/URC-94(V), as follows:

Nomenclature	Description
AN/URC-94(V)-1	Shock mounted transceiver
AN/URC-94(V)-2	Rack mounted transceiver

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Test, H.I. (High Impact), Shipboard Machinery Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic and Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-H-55258	Handset H-189( )/GR
MIL-A-55288	Antenna AS-1729/VRC

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection BY Attributes
MIL-STD-108	Definition Of And Basic Requirement For Enclosure For Electric And Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type 1 - Environmental And Type 1i-Internally Excited)
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

PUBLICATIONS

National Security Agency

CSESD-14 TSEC KY-58
CSESD-32 TSEC KY-58
CSESD-15 TSEC KY-75

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The AN/URC-94(V) shall be in accordance with the provisions of MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.2 Description. The AN/URC-94(V) shall be a modular solid state radio set providing simplex two way communications, compatible with communications security (COMSEC) equipments, aboard boats and craft and fixed shore installations. The AN/URC-94(V) shall be capable of operation in the AM, single sideband (SSB) with selectable upper sideband (USB) and lower sideband (LSB), and continuous wave (CW) modes of emission in the 1.5 through 29.9999 MHz frequency range and in the FM mode of emission in the 30.0 through 79.9999 MHz. The AN/URC-94(V) shall consist of a receiver-transmitter, HF and VHF antenna couplers and antennas, a control/interface unit, an amplifier-control to enable remote operation of the AN/URC-94(V) (except for frequency selection and tuning), a handset, and a mount. The HF antenna coupler shall operate in the 1.5 to 29.9999 MHz frequency range and the VHF coupler in the 30.0 to 79.9999 MHz frequency with the coupler automatically selected and tuned depending upon the frequency of operation. The control/interface unit shall automatically select the appropriate COMSEC equipment depending upon the mode of operation. The AN/URC-94(V) shall operate from sources of power 115/230 volts alternating current (AC) and 26.4 volts direct current (DC).

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3.3 Equipment composition. The radio sets shall be complete, ready for installation and operation with the exception of interconnecting cables. The radio sets shall be composed of the basic items specified in 3.3.1 and 3.3.2.

3.3.1 AN/URC-94(V)-1. The AN/URC-94(V)-1 radio set is intended for application aboard boats and craft and shall consist of the following:

a. One each receiver/transmitter, Radio RT-1230(V)-1/URC-94(V) including:
   1. One each inverter, power, 26.4 VDC, CV-3419/URC-94(V)
   2. One each modulator/demodulator, MD-1014/URC-94(V)
b. One each base, shock-mount, electrical equipment, MT-4883/URC-94(V)
c. One each coupler, antenna, HF CU-2184/URC-94(V)
d. One each antenna, VHF, AS-1729/VRC
e. One each antenna, 4.87 meters (m) (16 feet), fiberglass with feed through base
f. One each amplifier-control, C-10206/URC-94(V)
g. One each handset, H-189/GR
h. One each control/interface J-3641/URC-94(V)

3.3.2 AN/URC-94(V)-2. The AN/URC-94(V)-2 radio set is intended for fixed shore installation and shall consist of the following:

a. One each receiver/transmitter, radio RT-1230(V)-2/URC-94(V) including:
   1. One each inverter, power 26.4 VDC, CV-3419/URC-94(V)
   2. One each modulator/demodulator, MD-1014/URC-94(V)
b. One each coupler, antenna, HF, CU-2184/URC-94(V)
c. One each antenna, VHF, AS-1729/VRC
d. One each antenna, 4.87 m (16 feet), fiberglass with feed through base
e. One each amplifier-control C-10206/URC-94(V)
f. One each handset, H-189/GR
g. One each control/interface J-3641/URC-94(V)

3.4 COMSEC compatibility. The AN/URC-94(V) shall be compatible with communications security equipments in accordance with CSESD-14, CSESD-32, and CSESD-15.

3.4.1 Reliability. The AN/URC-94(V) shall have an upper test limit mean-time-between failure (MTBF) (θ_0 as defined in MIL-STD-781) of either radio set configuration specified in 3.3 of 1000 hours. The lower test limit MTBF (θ_1 as defined in MIL-STD-781) shall be 500 hours.

3.4.1.1 Equipment conditioning. Each production radio set shall be subjected to a 72-hour environmental burn-in as specified in 4.9.2.

3.4.2 Maintainability.

3.4.2.1 Qualitative maintainability requirements. The AN/URC-94(V) shall provide ease of maintenance, accessibility, and replacement of all modules and parts.

3.4.2.2 Quantitative corrective maintenance. The AN/URC-94(V), including built-in-test (BIT), shall have a mean-corrective-maintenance time (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenance time (M_{maxct}) (95th percentile) not exceeding 1.0 hour when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit (PC) boards) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. BIT circuits are included in the corrective maintenance time requirement. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

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3.4.3 Parts, materials, and processes. Parts, materials, and processes shall be selected using MIL-E-16400 as a guide to the maximum extent practicable.

3.4.3.1 Procedure for selection of electronic and electrical parts and material during design. Selection and prescreening of electronic and electrical parts and materials shall be in accordance with MIL-STD-965, Procedure I.

3.4.4 Safety. The safety requirements shall be in accordance with Requirement 1 of MIL-STD-454.

3.4.4.1 Leakage current. The leakage current of the equipment shall not exceed five milliamperes (mA).

3.4.5 Electromagnetic compatibility (EMC). The equipment shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class IA and IB equipments, with the following exceptions:

a. CE01: Broadband shall be exempted.
b. CS06: Shall be performed at 400 volts, five microseconds (μs)
c. RE02: Limit may be relaxed by 20 decibels (dB) at the transmitter fundamental frequency
d. RS03: Shall be performed at 10 volts per meter.

MIL-HDBK-241 may be used for guidance in electromagnetic interference (EMI) design of power supplies.

3.5 Receiver/transmitter RT-1230(V)/URC-94(V).

3.5.1 Physical characteristics.

3.5.1.1 Size. The overall size of the receiver/transmitter shall not exceed 49.53 centimeters (cm) (19.5 inches(in)) wide by 49.53 cm (19.5 in) deep by 20.32 cm (8.0 in) high for the RT-1230(V)-1/URC-94(V) and 49.53 cm (19.5 in) wide by 50.80 cm (20.0 in) deep by 22.86 cm (9.0 in) high for the RT-1230(V)-2/URC-94(V).

3.5.1.2 Weight. The weight of the receiver/transmitter shall not exceed 38.55 kilograms (kg) (85 pounds).

3.5.1.3 Modular construction. The receiver/transmitter shall use modular construction to the maximum extent practicable.

3.5.2 Environmental service conditions. The receiver/transmitter shall be capable of continuous reliable operation within the performance limits specified herein. The receiver/transmitter shall operate within tolerances throughout the specified tests without alignment or adjustment, other than accessible controls normally employed for operation of the equipment.

3.5.2.1 Temperature ranges. The receiver/transmitter shall maintain specified performances when exposed to the high and low temperatures specified in 3.5.2.1.1 and 3.5.2.1.2.

3.5.2.1.1 SSB (USB and LSB) and AM. The operating and non-operating temperatures shall be in accordance with Range 2 of MIL-E-16400.

3.5.2.1.2 FM and CW. The operating temperature range shall be from -28°C Celsius(C) to +50°C. The non-operating temperature range shall be from -62°C to +71°C.

3.5.2.2 Humidity. The receiver/transmitter shall maintain the specified performance when exposed to a relative humidity of 85 percent at an operating temperature of +40°C for both continuous and intermittent periods.

3.5.2.3 Salt fog. The receiver/transmitter shall comply with the salt fog requirement specified in MIL-E-16400.

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3.5.2.4 Shock. The receiver/transmitter shall be capable of withstanding the Grade A, lightweight Type A, Class II, shock test of MIL-S-901 when mounted on the MT-4883/URC-94(V) shock mount.

3.5.2.5 Vibration. The receiver/transmitter shall be capable of withstanding and shall conform to the performance requirements specified herein while subjected to the Type I vibration test of MIL-STD-167-1, when mounted on the MT-4883/URC-94(V) shock mount.

3.5.2.6 Inclination. The receiver/transmitter shall comply with the inclination requirement of MIL-E-16400.

3.5.3 Electrical design and construction.

3.5.3.1 Electrical power characteristics. The receiver/transmitter shall be designed to operate from primary power sources, as measured at the input of the equipment, as specified in 3.5.3.1.1 and 3.5.3.1.2.

3.5.3.1.1 AC power source. 115/230 volts AC ±10 percent single-phase 50 to 60 hertz (Hz).

3.5.3.1.2 DC power source. 26.4 volts DC nominal ±10 percent.

3.5.3.2 Reverse polarity protection. The receiver/transmitter shall include means to protect the equipment against accidental reversal of polarity when connected to the DC power source.

3.5.3.3 Fuses. All AC and DC electrical power inputs to the receiver/transmitter shall be protected by fast-blow type fuses of adequate rating.

3.5.3.4 Input electrical power selection. A three-position switch located in the rear panel shall be provided to select the type of electrical power input, either 115 volts AC, 230 volts AC, or DC.

3.5.3.5 Electrical power connectors. Separate connectors shall be provided for AC and DC power connectors. These connectors shall be located on the rear panel of the receiver/transmitter.

3.5.3.6 Bonding and grounding. Bonding and grounding shall be in accordance with the requirements of MIL-E-16400.

3.5.3.7 HF coupler power and control signals. The power and control signals shall be supplied to the CU-2184/URC-94(V) HF antenna coupler by the receiver/transmitter via a single multiconductor cable.

3.5.3.7.1 Primary power. The receiver/transmitter shall supply the primary input voltage required by the coupler at a voltage of 22 to 40 volts DC.

3.5.3.7.2 Output control signals. The receiver/transmitter shall provide the HF coupler with the following control signals to permit automatic operation:

a. Rechannel pulse
b. Transmit keyline

3.5.3.7.3 Input control signals. The receiver/transmitter shall accept the following control signals from the HF coupler to permit automatic monitor, operation monitor, and operation of the coupler:

a. Transmit keyline
b. Keyline enable
c. Tune power request
d. Tune/fault indication

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3.5.3.8 VHF antenna control signals. The receiver/transmitter shall provide the appropriate switching code signals required for tuning the AS-1729/VRC antenna.

3.5.3.9 Amplifier-control power. The receiver/transmitter shall provide the primary power required by the amplifier-control.

3.5.4 Mechanical design.

3.5.4.1 Mounting.

3.5.4.1.1 RT-1230(V)-1/URC-94(V). The receiver/transmitter RT-1230(V)-1/URC-94(V) shall be designed and constructed to be mounted into the base, shock mount, MT-4883/URC-94(V). The receiver/transmitter shall be easily removable from the mount with the use of two front surface fastening screws.

3.5.4.1.2 Base, shock mount, MT-4883/URC-94(V). The mount, MT-4883/URC-94(V) shall be a combination shock and vibration mount designed for deck or table mounting. The mount shall be designed and constructed to permit the receiver/transmitter to comply with the shock and vibration requirements of this specification. An air filter screen easily removable from the front, shall be integrated to the shock mount. The overall size of the shock mount shall not exceed 46.99 cm (18.5 in) wide by 43.18 cm (17.0 in) deep by 7.62 cm (3.0 in) high and its weight shall not exceed 7.71 kg (17.0 pounds).

3.5.4.1.3 RT-1230(V)-2/URC-94(V). The receiver/transmitter RT-1230(V)-2/URC-94(V) shall be designed and constructed to be rack or cabinet mounted and shall be provided with integral slides.

3.5.4.2 Enclosure. The degree of enclosure shall be splashproof in accordance with MIL-STD-108.

3.5.5 Controls, indicators, and connectors.

3.5.5.1 Controls. The receiver/transmitter shall be provided with operational controls as specified in TABLE I.

3.5.5.2 Indicators. The receiver/transmitter shall be provided with indicators as specified in TABLE II.

3.5.5.3 Connectors. The receiver/transmitter shall be provided with connectors as specified in TABLE III. All connectors which are not in continuous use shall be provided with a protective cap to prevent damage to the connector receptacle when the mating connector is not installed. The protective cap shall be affixed adjacent to the connector receptacle. The equipment shall be provided equipped with straight mating connectors.

3.5.6 Performance requirements.

3.5.6.1 General.

3.5.6.1.1 Frequency range. The receiver/transmitter shall be capable of emission and reception over the 1.5 to 79.9999 MHz frequency range.

3.5.6.1.2 Modes of operation. The receiver/transmitter shall be capable of emission/reception in the following modes of operation as defined in MIL-STD-188:

a. A2J: Continuous wave telegraphy (CW)
b. A3H: Telephone (AM compatible) with full carrier and upper sideband modulation (AM)
c. A3J: Single sideband suppressed carrier (SSB) selectable USB and LSB
d. F3: Frequency modulation (FM) (narrow and wideband)

3.5.6.1.3 Mode of operation restriction. An internal jumper system shall be provided and factory wired to inhibit operation over the 1.5 to 29.9999 MHz frequency range in the FM mode and to inhibit operation over the 30.0 to 79.9999 MHz frequency range in the AM, SSB, and CW modes.

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TABLE I. Controls.

Control	Location	Functional description
Transmit power switch	Front panel	Selects forward or reflected power for indication by the power meter
Receive tune switch	Front panel	When pressed, disables the squelch and applies a broadband noise signal to the receiver
Tune control	Front panel	Tunes the preselector to the operating frequency
Transmit audio control	Front panel	Controls transmit audio drive level (disabled in the FM mode)
Squelch control	Front panel	Selects one of two separate squelch modes of operation
Receive RF gain control	Front panel	Controls the sensitivity of the receiver
Receive audio control	Front panel	Controls audio level to speaker, earphones, or headset
Speaker switch	Front panel	Selects front panel speaker for monitoring audio signals
Keyline switch	Front panel	Selects local or both local and remote keying of the transmitter
VFO control	Front panel	Allows tuning of operating frequency ± 5 kHz from the frequency set on the front panel
Mode switch	Front panel	Controls application of primary power to transceiver and selects mode of operation
Frequency controls	Front panel	The six controls select the operating frequency in increments of 10 MHz, 1 MHz, 100 kHz, 10 kHz, 1 kHz and 100 Hz from left to right, respectively.
Power selector switch	Rear panel	A 3-position switch that selects the type of input power, either 115 VAC, 230 VAC, or 24 VDC

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TABLE II. Indicators.

Indicator	Location	Functional description
Power meter	Front panel	In receive mode, indicates the received signal strength in S units; in transmit mode, indicates forward or reflected power as selected by transmit power switch
Reset indicator	Front panel	When illuminated indicates that protect circuitry has activated

TABLE III. Connectors.

Connector	Location	Description
Phone jack	Front panel	Connector for headphones
CW key jack	Front panel	Connector for hand key used for CW transmission
Audio connector	Front panel	Connector for microphone or headset
Remote audio connector	Rear panel	Interfaces remote audio accessories with a balanced 600 ohms line for narrow and wideband transmit and receive, keyline, DC power, and ground.
Antenna control connector	Rear panel	Interfaces the CU-2184/URC-94(V) and the AS-1729/VRC with the transceiver
DC input connector	Rear panel	Used to supply DC voltage to the transceiver when the +24 VDC power inverter is installed
AC input	Rear panels	Used to supply AC voltages to the transceiver
HF antenna connector	Rear panel	UHF type
VHF antenna connector	Rear panel	BNC type
Ground terminal	Rear panel	Stud with wing nut for ground connection

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3.5.6.1.4 Tuning. Frequency selection over the complete frequency range of the radio set shall be accomplished in 100 Hz increments by means of synthesizer controls. In addition, a means shall be provided for unlocking the frequency synthesizer to allow continuous tuning by means of a variable frequency oscillator (VFO) vernier frequency control. This vernier control shall permit continuous tuning of the receiver/transmitter at any frequency up to ± 5 kHz from the synthesizer frequency setting. Final tuning of the receiver/transmitter shall be accomplished by adjusting a tune control for maximum meter reading or maximum receiver-noise. A pushbutton switch on the front panel shall be provided to activate a tuning signal.

3.5.6.1.5 Automatic antenna matching. The receiver/transmitter shall automatically select the HF antenna coupler CU-2184/URC-94(V) or the VHF antenna AS-1729/VRC depending upon the frequency selected on the synthesizer frequency selection controls and shall provide appropriate control and tune signals to the selected coupler for automatically matching the output of the receiver/transmitter to the associated antenna.

3.5.6.1.6 Coupler tune signal. When the HF coupler CU-2184/URC-94(V) is selected, the receiver/transmitter shall emit an audible tone during the tuning cycle of the coupler.

3.5.6.1.7 Fault alert. When the HF coupler CU-2184/URC-94(V) is selected, the receiver/transmitter shall emit an interrupted audible tone when the coupler is unable to match the receiver/transmitter to the antenna because of a high voltage standing wave ratio (VSWR) or in case of malfunction.

3.5.6.1.8 Frequency stability. With the VFO OFF, the frequency stability of the receiver/transmitter shall be plus or minus one part in one million in the SSB/AM/CW, and narrowband FM modes. In the wideband FM mode, the transmit carrier shall be within ± 1500 Hz of the frequency indicated by the frequency selection controls.

3.5.6.2 Transmitter.

3.5.6.2.1 Output power. The output power of the transmitter shall be within ± 1 dB of the specified nominal powers over the applicable frequency range for the following modes of operation:

a. SSB: 100 watts peak-envelope-power (PEP) nominal, (100 watts average) single tone modulation
b. AM compatible: 100 watts PEP nominal (25 watts average carrier)
c. CW: 100 watts average nominal
d. FM: 50 watts average nominal

3.5.6.2.2 Output impedance. The transmitter shall deliver the specified output powers into a matched load impedance of 50 ohms nominal.

3.5.6.2.3 Duty cycle. The transmitter shall deliver the specified output powers on a continuous duty cycle basis.

3.5.6.2.4 Overload protection. The transmitter shall be provided with means of protection against the effects of open and short circuit conditions.

3.5.6.2.5 Carrier suppression. In the SSB mode of operation the carrier shall be suppressed at least 50 dB below the rated PEP power output.

3.5.6.2.6 Undesired sideband suppression. In the SSB mode of operation the undesired sideband shall be suppressed at least 50 dB at 1 kHz relative to the rated PEP power.

3.5.6.2.7 Harmonic suppression. All harmonics shall be suppressed at least 35 dB relative to the rated PEP power.

3.5.6.2.8 Spurious emissions. In the SSB mode of operation spurious emission shall be suppressed at least 40 dB on 99 percent of the channels relative to the full rated power output.

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3.5.6.2.9 Intermodulation. In the SSB mode of operation intermodulation products shall be at least 25 dB below one of two equal tones that drive the transmitter to full rated power output.

3.5.6.2.10 Automatic level control (ALC). The transmitter shall be provided with an ALC circuit. The ALC circuit shall enable the transmitter to maintain the power output specified herein over the 1.5 to 79.9999 MHz frequency range.

3.5.6.2.11 Average carrier control (ACC). The transmitter shall be provided with an ACC circuit. In the AM mode of operation this ACC circuit shall enable the transmitter to maintain a constant reinserted carrier level.

3.5.6.2.12 Audio inputs. Audio inputs shall be as follows:

a. Local inputs. The transmitter shall be capable of operating from audio signals generated by a standard dynamic microphone, carbon microphone, and an H-189/U field handset.
b. Remote audio input. The transmitter shall be capable of operating from remote audio inputs of -20 dBm to +10 dBm, 600 ohms balanced ± 10 percent. For wideband FM, the transmitter shall be capable of operation from a remote audio input of 0 dBm nominal, 600 ohms unbalanced.

3.5.6.2.13 FM deviation. The FM deviation shall be 8.0 kHz ± 2 kHz.

3.5.6.2.14 FM deviation limiting in the narrowband FM mode. FM deviation limiting circuitry shall be provided such that with a 14 dB increase above standard audio input voltage, the deviation shall not increase more than 3 dB.

3.5.6.2.15 Narrowband FM audio frequency response. The frequency response between 500 and 3000 Hz shall be within ± 3 dB of the response at 1000 Hz.

3.5.6.2.16 Wideband FM audio frequency response. The frequency response between 10 Hz and 17.0 kHz shall be within ± 6 dB of the response at 1000 Hz.

3.5.6.2.17 FM, 150 Hz tone transmission. A 150 Hz tone oscillator shall be provided for use in FM subcarrier squelch control. This 150 Hz tone shall frequency modulate the transmitter with a resulting deviation of 3.0 kHz ± 0.5 kHz. This function shall be automatically disabled in the wideband FM mode of operation.

3.5.6.2.18 Sidetone. Sidetone shall be provided in the CW mode of operation. The level of the side tone shall be adjustable up to one watt maximum output.

3.5.6.3 Receiver.

3.5.6.3.1 Input impedance. The input impedance shall be 50 ohms nominal.

3.5.6.3.2 Sensitivity. The sensitivity of the receiver measured at the 600 ohms remote audio output shall comply with the following requirements over the applicable frequency range for each mode of operation.

3.5.6.3.2.1 SSB. An unmodulated input signal of 0.5 microvolts (μV) referenced into 50 ohms (soft microvolts) 1 kHz in the desired sideband shall produce an output signal plus noise-to-noise ratio (S+N/N) of at least 10 dB.

3.5.6.3.2.2 FM (narrowband). An input signal of 0.75 microvolts referenced into 50 ohms (soft microvolts) frequency modulated with a 1000 Hz audio signal to a deviation of ± 8.0 kHz shall produce an output SINAD of at least 10 dB.

3.5.6.3.2.3 FM (wideband). An input signal of 2.5 microvolts referenced into 50 ohms (soft microvolts) frequency modulated with a 1000 Hz audio signal to a deviation of ± 8.0 kHz shall produce an output signal-plus-noise-plus distortion to noise plus distortion (SINAD) of at least 10 dB.

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3.5.6.3.3 Intermediate frequency (IF) selectivity. The selectivity of the IF filters shall comply with the requirements of 3.5.6.3.3.1 through 3.5.6.3.3.3.

3.5.6.3.3.1 SSB. Relative to a peak in the passband, the amplitude response (at 25°C) shall not exceed -6 dB from 350 to 3150 Hz. Beyond 4800 Hz the amplitude response shall be at least -55 dB relative to a peak in the passband.

3.5.6.3.3.2 AM/FM. Relative to a peak in the passband, the amplitude response (at 25°C) shall not exceed -6 dB from ± 18 kHz on center frequency. Beyond ± 40 kHz on center frequency, the amplitude response shall be at least -60 dB relative to a peak in the passband.

3.5.6.3.3.3 CW. Relative to a peak in the passband, the amplitude response (at 25°C) shall not exceed -3 dB from ± 160 Hz of the center frequency. The center frequency is the dial frequency plus 1000 Hz. Beyond ± 1000 Hz of the center frequency, the amplitude response shall be at least -60 dB relative to a peak in the passband.

3.5.6.3.4 FM audio frequency response (wideband). Audio frequency response of the receiver shall be within ± 6 dB of a 1000 Hz reference level from 10 Hz to 17 kHz when measured at the wideband output.

3.5.6.3.5 FM audio frequency response (narrowband). Audio frequency response shall be within ± 3 dB of 1000 Hz reference level over the 500 to 3000 Hz frequency range when measured at the narrowband output.

3.5.6.3.6 Audio output power.

3.5.6.3.6.1 Speaker. The output power to the internal speaker shall be three watts maximum at five percent maximum distortion.

3.5.6.3.6.2 Remote output. The output power shall be 0 dBm into 600 ohms.

3.5.6.3.7 Audio output impedance. The audio output impedance shall be 600 ohms balanced ± 20 percent. The wideband output impedance shall be 600 ohms unbalanced.

3.5.6.3.8 Image rejection. The rejection of image signals shall be at least 70 dB when the receiver is conditioned to receive a desired RF signal of the specified sensitivity at any frequency between 1.5 and 79.9999 MHz.

3.5.6.3.9 Intermediate frequency (IF) rejection. Signals at the intermediate frequencies shall be rejected at least 70 dB when the receiver is conditioned to receive a signal at the specified sensitivity at any frequency between 1.5 and 79.9999 MHz.

3.5.6.3.10 In-band intermodulation. In the SSB mode of operation, 3rd order intermodulation products at the 600 ohms narrowband audio output shall be at least 20 dB below either tone of a two equal amplitude in channel radio frequency signal at the receiver input. This requirement shall be conformed to with an input signal of up to 10 millivolts (mV) per tone.

3.5.6.3.11 Cross modulation. In the SSB mode of operation, with an undesired signal at a frequency as close as three percent from the desired frequency, 30 percent modulated and 60 dB greater than a desired signal, the cross modulation shall be at least 20 dB below a desired signal of up to 1 mV (soft microvolts).

3.5.6.3.12 Blocking (desensitization). The output S+N/N ratio due to a desired signal 1 mV (soft microvolt) shall not be degraded more than 6 dB by an undesired signal of one millivolt (soft microvolts) as close as one percent from the desired frequency.

3.5.6.3.13 AM rejection. In the FM mode of operation the output due to a signal amplitude modulated 30 percent by a 1.0 kHz audio signal shall be at least 40 dB below the output due to a signal frequency modulated with a 1.0 kHz audio signal to a deviation of ± 8 kHz.

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3.5.6.3.14 Internal spurious. Ninety-nine percent of the internal spurious shall be below 0.5 μV (soft microvolt) equivalent input signal at the antenna input.

3.5.6.3.15 External spurious response. In the SSB mode of operation spurious signals outside the receiver passband shall be rejected at least 70 dB relative to a 0.5 μV (soft microvolt) desired signal as measured at the 600 ohms remote audio output, over the 1.5 to 29.9999 MHz frequency range.

3.5.6.3.16 Automatic gain control (AGC) characteristics.

3.5.6.3.16.1 Attack time constant. The attack time constant shall be 10 milliseconds (ms) nominal in the SSB, AM, and FM modes of operation.

3.5.6.3.16.2 Release time constant. The SSB release time constant shall be within three to five seconds. The FM time constant shall be 100 ms.

3.5.6.3.16.3 Threshold. The AGC threshold shall be adjustable from 3 μV to 12 μV (soft microvolts).

3.5.6.3.16.4 Slope. Except in the FM mode of operation, with the AGC threshold set at 3 μV, an input signal variation from 5.0 μV to 500 mV shall not cause a change in audio output greater than 12.0 dB.

3.5.6.3.17 Antenna radiation. The antenna radiation at the antenna input shall be less than 100 picowatts into 50 ohms.

3.5.6.3.18 Overload protection. The receiver shall be provided with an overload protection circuit which shall be activated when the received signal at the antenna input is within three to five volts rms.

3.6 Antenna coupler CU-2184/URC-94(V).

3.6.1 Physical characteristics.

3.6.1.1 Size. The overall size of the antenna coupler shall not exceed 35.56 cm (14.0 in) wide by 48.26 cm (19.0 in) deep by 21.59 cm (8.5 in) high.

3.6.1.2 Weight. The weight of the antenna coupler shall not exceed 13.6 kilograms (kg) (30 pounds).

3.6.2 Environmental service conditions. The antenna coupler shall be capable of reliable continuous operation within the performance limits specified herein. The antenna coupler shall operate within tolerances throughout the specified tests without alignment or adjustment other than accessible controls normally employed for operation of the equipment.

3.6.2.1 Temperature ranges. The operating and non-operating temperature range shall be in accordance with Range 2 of MIL-E-16400.

3.6.2.2 Humidity. Humidity requirements shall be in accordance with MIL-E-16400.

3.6.2.3 Salt fog. Salt fog requirements shall be in accordance with MIL-E-16400 for exposed equipment.

3.6.2.4 Shock. The shock requirements shall be in accordance with MIL-E-16400.

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3.6.2.5 Vibration. The antenna coupler shall be capable of withstanding and shall conform to the performance requirements specified herein while subjected to the Type I vibration test of MIL-STD-167-1.

3.6.2.6 Inclination. The inclination requirements shall be in accordance with MIL-E-16400.

3.6.2.7 Dust. The equipment shall comply with the performance requirements specified herein after being subjected to Test Method 510, Procedure 1, of MIL-STD-810.

3.6.2.8 Rain. The equipment shall comply with the performance requirements specified herein after being subjected to Test Method 506, Procedure 1, of MIL-STD-810.

3.6.2.9 Orientation. The coupler shall be capable of storage and operation in any orientation.

3.6.2.10 Siting. The coupler shall be installed at the base of the associated antenna. The separation between the coupler and the associated RT-1230(V)/URC-94(V) receiver/transmitter shall not exceed 76.2 m (250 feet).

3.6.2.11 Enclosure. The degree of enclosure shall be watertight and airtight in accordance with MIL-STD-108.

3.6.3 Electrical requirements.

3.6.3.1 Primary power source. The primary power input shall be derived from, and shall be compatible with, the RT-1230(V)/URC-94(V) receiver/transmitter. The primary input power to the coupler shall be 22 to 40 volts DC, and power consumption during the tuning cycle shall not exceed 30 watts average, 200 watts peak.

3.6.3.2 Control signals.

3.6.3.2.1 Input control signals. The coupler shall accept the following control signals from the RT-1230(V)/URC-94(V) receiver/transmitter to permit automatic operation:

a. Rechannel pulse
b. Transmit keyline

3.6.3.2.2 Output control signals. The coupler shall provide the RT-1230(V)/URC-94(V) receiver/transmitter with the following control signals to permit automatic and operation monitor and operation of the coupler:

a. Transmit keyline
b. Keyline enable
c. Tune power request
d. Tune/fault indication

3.6.3.3 Power and control signals. The power and control signals between the coupler and RT-1230(V)/URC-94(V) shall be capable of being supplied by means of a single multiconductor cable. This cable shall not be supplied with the equipment.

3.6.3.4 Grounding stud. A suitable grounding stud shall be provided on the case near the antenna insulator.

3.6.4 Mechanical.

3.6.4.1 Mounting. The coupler shall be capable of being mounted in the operating environment so that the separation between the RF connection at the base of the antenna and the RF terminal on the coupler is 30.48 cm (12 in) nominal. This connection shall be made with a suitable copper conductor. The coupler shall be capable of installation on a flat surface having a minimum width of 32.63 cm (12.85 in) and a minimum length of 43.81 cm (17.25 in) without the requirement for resilient shock and vibration mounts.

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3.6.4.2 Cooling. The coupler shall provide the specified performance without the requirement for external cooling facilities.

3.6.4.3 Hardware. All screws, bolts, nuts, and other hardware shall be of austenitic corrosion-resistant steel, except where high RF fields require the use of brass.

3.6.4.4 Connectors. A single power and control cable connector and a single RF input connector shall be located at the opposite end of the coupler from the antenna connector. All external mating plugs shall be provided with the coupler. All externally mounted connector receptacles shall be provided with a protective cap to prevent damage to the connector during shipment and storage when the mating connector is not installed. A single porcelain or ceramic antenna connector shall be provided at one end of the coupler. The connector shall be capable of withstanding the high RF voltages present during operation at full rated output. The connector shall be provided with a suitably threaded stud to accommodate the antenna lead.

3.6.4.5 Hazard warning. A warning sign shall be permanently affixed to the antenna coupler near the antenna terminal stating the hazard involved with contacting the antenna terminal.

3.6.5 Performance requirements.

3.6.5.1 General requirement. The HF antenna coupler CU-2184/URC-94(V) shall be designed to be compatible with and shall be operated automatically by the receiver/transmitter R-1230(V)/URC-94(V).

3.6.5.2 Frequency range. The HF antenna coupler shall operate over the 1.5 to 29.9999 MHz frequency range.

3.6.5.3 Method of tuning.

3.6.5.3.1 Description. Tuning of the HF antenna coupler shall be automatic. Upon receipt of a rechannel pulse from the associated receiver/transmitter, or upon initial power-on condition, the antenna coupler tuning elements shall proceed to the HOME position. During this portion of the tuning cycle, the transmitter shall be inhibited from transmitting. Also during this portion of the cycle, and after the tuning elements have reached the HOME position in an unkeyed condition, the tuning elements shall be bypassed as specified in 3.6.5.10.2. If at any time after receiving a rechannel pulse the coupler receives a momentary keyline ground from the associated receiver/transmitter, the coupler shall lock-up the keyline and keep it grounded. It shall also produce a constant signal on the tune/fault indication line to enable a tune tone in the receiver/transmitter. After the tuning elements have reached the HOME position, the transmit inhibit signal shall be released. After receipt of a keyline ground, the coupler shall disable the bypass function as specified in 3.6.5.10.2 and shall enable the tune power circuits of the associated receiver/transmitter. When RF tune power is received, the coupler shall automatically tune the antenna to a VSWR as specified herein and within the specified time. When this tuning is accomplished, the coupler shall release the keyline ground and shall disable the signal on the tune/fault indication line. Normal transmission from the transmitter shall then be allowed, and VSWR protection circuitry shall be enabled.

3.6.5.3.2 High VSWR protection. In the event of a high reflected power (high VSWR) the transmitter shall be inhibited from transmitting, and an intermittent signal on the tune/fault indication line shall be produced by the coupler to cause an intermittent tone in the associated receiver/transmitter. Upon receipt of a rechannel pulse from the receiver/transmitter, this intermittent tone shall be removed and the tuning elements of the coupler shall proceed to the HOME position to start a new tuning sequence.

3.6.5.4 Power capability. The coupler shall be capable of safely handling 100 watts plus 1 dB peak and average RF power over the 1.5 to 29.9999 MHz frequency range. The coupler power handling capability shall be compatible with the power output of the receiver/transmitter RT-1230(V)/URC-94(V) in the 1.5 to 29.9999 MHz frequency range.

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3.6.5.5 Duty cycle. The coupler shall be capable of safe and reliable continuous operation at full rated power over the 1.5 to 29.9999 MHz frequency range under the environmental extremes specified herein, upon completion of final tuning procedures.

3.6.5.6 Antennas. The HF antenna coupler shall be capable of automatically matching the following antennas to the output of the associated receiver/transmitter:

a. 7.62 m to 10.66 m (25- to 35-foot) whip antenna over the 1.5 to 29.9999 MHz frequency range
b. 22.86 m to 45.72 m (75- to 150-foot) long wire antenna over the 1.5 to 29.9999 MHz frequency range with the use of a long wire antenna adapter
c. 4.57 m (15-foot) whip antenna over the 2.0 to 29.9999 MHz frequency range

3.6.5.7 VSWR. The HF coupler shall automatically match any antenna impedance within its capabilities to a 50-ohm transmission line with a VSWR not greater than 1.5:1.

3.6.5.8 Tuning time. Tuning time shall not exceed 15 seconds. Tuning time is the interval of time between the time a tune signal is activated with the coupler in the HOME position and the time at which automatic tuning of the coupler elements is completed at the new operating frequency.

3.6.5.9 Tune RF power. During the tune cycle the coupler shall operate with a RF tune input forward power of 10 watts minimum, supplied from the associated receiver/transmitter.

3.6.5.10 Modes of operation

3.6.5.10.1 Normal operation. During normal operation, the coupler shall match the antennas specified herein to the 50-ohm coaxial cable from the output of the associated receiver/transmitter.

3.6.5.10.2 Bypass operation. During bypass operation, the coupler shall connect the antenna directly to the 50-ohm coaxial cable from the associated receiver/transmitter with no detrimental effects to the RF signal over those that would be present if no coupler was used.

3.7 Antenna AS-1729/VRC. The antenna AS-1729/VRC shall comply with the requirements of MIL-A-52288.

3.8 Amplifier/control C-10206/URC-94(V).

3.8.1 Description. The amplifier/control unit C-10206/URC-94(V) shall provide for remote operation of the RT-1230(V)/URC-94(V) receiver/transmitter with up to 30.48 m (100 feet) separation between the units. The control unit shall provide a microphone input, a loudspeaker, transmitter keyline, audio output control, and ON/OFF power control of the associated RT-1230(V)/URC-94(V).

3.8.2 Size and weight. The overall size of the audio control unit shall not exceed 21.59 cm (8.5 in) wide by 13.97 cm (5.5 in) deep by 15.24 cm (6.0 in) high and its weight shall not exceed 1.81 kg (4.0 pounds).

3.8.3 Temperature range. The operating and non-operating temperature range shall be in accordance with Range 2 of MIL-E-16400.

3.8.4 Humidity. Humidity requirements shall be in accordance with MIL-E-16400.

3.8.5 Salt fog. Salt fog requirements shall be in accordance with MIL-E-16400.

3.8.6 Shock. The amplifier/control shall be capable of withstanding the Grade A, lightweight Type A, Class II, shock test of MIL-S-901.

3.8.7 Vibration. The amplifier/control shall be capable of withstanding the Type I vibration test of MIL-STD-167-1.

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3.8.8 Inclination. The amplifier/control shall comply with the inclination requirements of MIL-E-16400.

3.8.9 Enclosure. The degree of enclosure shall be splash-proof in accordance with MIL-STD-108.

3.9 Control/interface J-3641/URC-94(V). The control/interface shall be electrically and mechanically compatible with the RT-1230(V)/URC-94(V) and the communication security equipments specified herein. The control/interface shall automatically connect the input and output of the appropriate communication security equipment to the appropriate output and input of the receiver/transmitter, respectively, depending upon the selected receiver/transmitter frequency. This control/interface shall conform to the environmental requirements specified in 3.8.3 through 3.8.8.

3.9.1 Size and weight. The overall size of the control/interface shall not exceed 20.32 cm (8.0 in) wide by 11.43 cm (4.5 in) deep by 8.89 cm (3.5 in) high and shall not weigh more than 1.81 kg (4.0 pounds).

3.9.2 Control/interface enclosure. The degree of enclosure shall be splashproof in accordance with MIL-STD-108.

3.10 Handset. The handset shall be a military standard handset H-189/GR in accordance with MIL-H-55258.

3.11 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.3 General inspection. Inspection shall be in accordance with MIL-E-16400 except as otherwise specified herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
c. Environmental inspection (see 4.4.3)
d. Reliability tests (see 4.5)
e. Maintainability demonstration (see 4.6)
f. Inspection of preparation for delivery (see 4.10)

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4.3 First article inspection. Unless otherwise specified (see 6.2) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV in accordance with the First Article Inspection paragraph of MIL-E-16400.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be in accordance with MIL-E-16400 and TABLE IV and shall be performed on every equipment offered for delivery.

4.4.2 Production control inspection. Production control inspection shall be in accordance with MIL-E-16400 and TABLE IV. Sampling shall conform to the inspection procedures of MIL-STD-105, using the special inspection levels. The inspection level shall be S-4 for normal, tightened, and reduced inspection with an acceptable quality level (AQL) of 4.0 percent. Production control inspection shall be performed on equipments that have passed the production inspection.

TABLE IV. First article (FA), production (P), production control (PC), and environmental (E) inspections.

Inspection or test	Requirement paragraph	Inspection type			
		FA	P	PC	E
COMSEC compatibility	3.4	X			
Reliability	3.4.1				
Equipment conditioning	3.4.1.1		X		
Maintainability	3.4.2				
Parts, materials, and processes	3.4.3	X	X		
Workmanship	3.11				
Safety	3.4.4	X			
Leakage current	3.4.4.1	X		X	
EMC	3.4.5	X			X
Size	3.5.1.1, 3.5.4.1.2; 3.6.1.1, 3.8.2, 3.9.1	X	X		
Weight	3.5.1.2, 3.5.4.1.2, 3.6.1.2, 3.8.2, 3.9.1	X	X		
Temperature ranges	3.5.2.1, 3.6.2.1, 3.8.3	X			X
Humidity	3.5.2.2, 3.6.2.2, 3.8.4	X			X
Salt fog	3.5.2.3, 3.6.2.3, 3.8.5	X			X
Shock	3.5.2.4, 3.6.2.4, 3.8.6	X			X
Vibration	3.5.2.5, 3.6.2.5, 3.8.7	X			X
Inclination	3.5.2.6, 3.6.2.6, 3.8.8	X			X
Enclosure	3.5.4.2, 3.6.2.11, 3.8.9, 3.9.2	X			
Cooling	3.6.4.2	X			
Electrical power characteristics	3.5.3.1, 3.6.3.1	X			
Frequency range	3.5.6.1.1, 3.6.5.2	X	X		
Modes of operation	3.5.6.1.2, 3.6.5.10,	X	X		
restriction	3.5.6.1.3	X	X		
Automatic antenna matching	3.5.6.1.5	X	X		
Coupler tune signals	3.5.6.1.6				
Fault alarm	3.5.6.1.7	X	X		
Frequency stability	3.5.6.1.8	X	X		X
RF output power	3.5.6.2.1, 3.6.5.4	X	X		
Output impedance	3.5.6.2.2	X		X	
Duty cycle	3.5.6.2.3, 3.6.5.5	X			X

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TABLE IV. First article (FA), production (P), production control (PC), and environmental (E) inspections. (Continued)

Inspection or test	Requirement paragraph	Inspection type			
		FA	P	PC	E
Overload protection	3.5.6.2.4, 3.5.6.3.18	X	X		
Carrier suppression	3.5.6.2.5	X		X	X
Undesired sideband suppression	3.5.6.2.6	X		X	X
Harmonic suppression	3.5.6.2.7	X		X	X
Spurious emissions	3.5.6.2.8, 3.5.6.3.14, 3.5.6.3.15	X		X	X
Intermodulation	3.5.6.2.9, 3.5.6.3.10	X		X	X
Automatic level control	3.5.6.2.10	X	X		
Average carrier control	3.5.6.2.11	X	X		
Audio inputs	3.5.6.2.12	X			
Audio response	3.5.6.2.15, 3.5.6.2.16, 3.5.6.3.4	X		X	
Input impedance	3.5.6.3.1	X			
Sensitivity	3.5.6.3.2	X	X		X
IF selectivity	3.5.6.3.3	X		X	X
Audio output power	3.5.6.3.6	X		X	X
Audio output impedance	3.5.6.3.7	X			
FM deviation	3.5.6.2.13	X		X	
FM deviation limiting	3.5.6.2.14	X		X	
FM, 150 Hz tone transmission	3.5.6.2.17	X	X		
Sidetone	3.5.6.2.18	X	X		
Image rejection	3.5.6.3.8	X		X	X
IF rejection	3.5.6.3.9	X		X	X
Crossmodulation	3.5.6.3.11	X		X	X
Blocking	3.5.6.3.12	X		X	X
AM rejection	3.5.6.3.13	X		X	X
AGC characteristics	3.5.6.3.16	X	X		
Antenna radiation	3.5.6.3.17	X	X		
Dust	3.6.2.7	X			X
Rain	3.6.2.8	X			X
Overload protection	3.5.6.3.18	X		X	
High VSWR protection	3.6.5.3.2	X	X		
VSWR	3.6.5.7	X	X		
Tuning time	3.6.5.8	X		X	
Tune RF power	3.6.5.9	X		X	
Amplifier/control compatibility	3.8.1	X		X	
Control/interface compatibility	3.9	X		X	

4.4.2.1 Rejection of inspection lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection. Environmental inspection shall be in accordance with MIL-E-16400 and TABLE IV, and shall be performed on units which have been subjected to and passed production inspection and production control inspection.

4.4.3.1 Sampling for environmental inspection shall be as follows:

a. One of the first five production units
b. One of the next 20 production units
c. One of the next 100 units
d. One of the final lot if more than 50 units are produced

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4.4.3.2 Should a unit fail environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government quality assurance representative the results thereof and provide details of the corrective action taken to correct units which were manufactured under the same conditions, with the same materials, processes and so forth.

4.4.3.3 Sample units which have been subjected to and passed production control or environmental inspection, or both, may be accepted on the contract provided that any damage or deterioration resulting from those tests is corrected and the units are resubjected to and pass the production inspection.

4.5 Reliability testing. A sample of at least three AN/URC-94/(V)-1 radio sets (excluding antenna and H-189/U handset) shall be selected from the first month's production for reliability qualification testing in accordance with Test Plan IV of MIL-STD-781. Environmental and operational duty cycles shall be as specified herein. The reliability test shall be successfully completed prior to delivery of any equipments under the contract.

4.5.1 Temperature cycling. Temperature cycling between 0°Celsius (C) (+3°C, -0°C) and +50°C (+0°C, -3°C) shall be performed throughout the test with the following constraints:

a. Time duration at the stabilized high and low temperatures shall be approximately three hours.
b. During transition between temperature extremes, the average rate of chamber temperature change shall not be less than 5°C per minute.

4.5.2 Vibration. A 4 Hz to 50 Hz to 4 Hz vibration sweep shall be performed throughout the reliability test with vibratory displacement (table amplitude) as defined by Table I of MIL-STD-167-1. The duration of each complete sweep shall be approximately 15 minutes.

4.5.3 Operating duty cycle. Equipment shall be non-operating (input power removed) during the final 30 minutes of each stabilized low temperature period. During all remaining test periods, the equipment shall be operating in accordance with TABLE V with a transmit/receive duty cycle of five minutes transmit and 10 minutes receive. The schedule of TABLE V shall be repeated as necessary.

TABLE V. Operating schedule.

Day	Transmit mode	Receive mode
1	CW: 100 watts average power keyed at 32 band	CW: Keyed at 32 band, one microvolt level
2	AM: 25 watts carrier modulated 30 to 50 percent with a 1000 Hz tone	AM: Carrier modulated 30 to 50 percent at 1000 Hz, one microvolt level
3	SSB: 100 watts PEP, 1000 Hz tone	SSB: Carrier modulated at 1000 Hz, one microvolt level
4	FM: 50 watts average, modulated at 1000 Hz tone	FM: Carrier modulated at 1000 Hz, one microvolt level

Frequencies shall be selected and alternated at least daily in a manner which will exercise the equipment throughout the specified frequency range. Only actual operating time shall be accumulated for purposes of establishing an accept/reject decision.

4.5.4 Input power. Input power to test samples shall be 230 VAC, 60 Hz.

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4.5.5 Failure. The procuring activity shall be notified within two working days of any failure during reliability testing. Classification of failures shall be made by the procuring activity in accordance with the Failure Actions paragraph of MIL-STD-781.

4.5.6 Rejected equipment. In the event equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for such rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and retrofitted in the test units, the reliability qualification test shall be reperformed as specified herein.

4.5.7 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reconditioning and reinspection as required and shall be subjected to the production tests of 4.4.1 to assure that it conforms to the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it conforms to all terms and conditions of the criteria for acceptance.

4.6 Maintainability demonstration.

4.6.1 Maintainability (equipment) demonstration. The contractor shall perform a maintainability and BIT/diagnostics demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment corrective maintenance (M_{ct})/BIT/diagnostics requirements are fulfilled.

4.6.2 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PC boards), and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471) and assigned numbers of 1 to 100. The following information shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

Faults of BIT functions shall be included as corrective maintenance tasks for the maintainability demonstration. The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 50 tasks for the demonstration.

4.6.3 Accept/reject criteria.

4.6.3.1 Corrective maintenance. The accept/reject criteria for the demonstration of the corrective maintenance time for the demonstration shall be as specified in TABLE VI.

TABLE VI. Acceptance criteria.

	Acceptance criteria	
	M_{ct}	M_{maxct}
Acceptance level	20	0
Sample size	50	50

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in the acceptance table corresponding to each index for the specified confidence level. The duration

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of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made when an accept decision is made for both M_{ct} and M_{maxct}. Accept/reject criteria for preventive maintenance shall be the capability of being able to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.6.4 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the equipment shall not interfere with on-line operation.

4.6.5 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.6.6 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.6.7 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.6.8 Rejection. Failure to conform to any of the requirements specified herein shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.6.9 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to any delivery.

4.7 Test facilities, chambers, and apparatus. Test facilities, chambers, and apparatus shall be in accordance with MIL-E-16400.

4.8 Method of performance of tests. Tests shall be performed in accordance with MIL-E-16400.

4.8.1 Standard test conditions. Unless otherwise specified, all performance tests shall be performed under the following standard test conditions:

a. Temperature: $20^{o}C$ to $30^{o}C$
b. Humidity: 85 percent or less
c. Input primary power (measured at the input to the radio) 115 volts AC, 60 Hz, single-phase or 26.4 volts DC
d. Shock and vibration: None
e. Altitude: Mean-sea-level
f. Transmitter dummy load: 50 ohms resistive
g. Audio input signal source impedance: 600 ohms
h. Receiver RF input signal source impedance: 50 ohms
i. Receiver remote audio output: Terminated with 600 ohms
j. Antenna load: 50 ohms resistive in series with 100 picofarads (pF)

4.8.1.1 Control settings. The control settings shall be as follows:

a. VFO: Off (in detent)
b. Speaker: On
c. Receiver RF gain: Maximum
d. Receiver audio: Mid-range
e. Transmitter audio: Auto (in detent)
f. Squelch: Fully counter-clockwise (but not in tone detent)
g. Keyline: Local

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4.8.1.2 Tuning. The preselector shall be tuned to the operating frequency by using the broadband noise signal which is activated by depressing the receiver tune signal switch, and by adjusting the tune control for maximum receiver noise.

4.9 Test methods. Test methods shall be in accordance with MIL-E-16400, except as otherwise specified herein.

4.9.1 Leakage current. A warning plate shall be affixed to the equipment reading as follows:

WARNING: THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.9.1.1 Equipment connections. Leakage current shall be measured on equipment in its normal operating configuration. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened. If the equipment must be touched during the test, personal protective devices, (for example, rubber gloves, insulating tools, or knobs), shall be used for protection against electric shock. The equipment shall be connected as shown in FIGURE 1. Measurement under OC power source condition is not required.

4.9.1.2 Measurement. The leakage current shall be determined by the voltage-drop method. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed five percent. A True root-mean-square (rms) voltmeter shall be used. The probe shall be connected to the equipment enclosure and associated unit enclosures and measurements taken for every combination of switch positions available in the test diagram. The probe shall also be connected to all external conducting parts such as connector housings, case, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. Equipments having more than one mode of operation, such as transmit and receive, shall be tested in each mode. The open safety ground conductor shall be reconnected immediately after the test is completed.

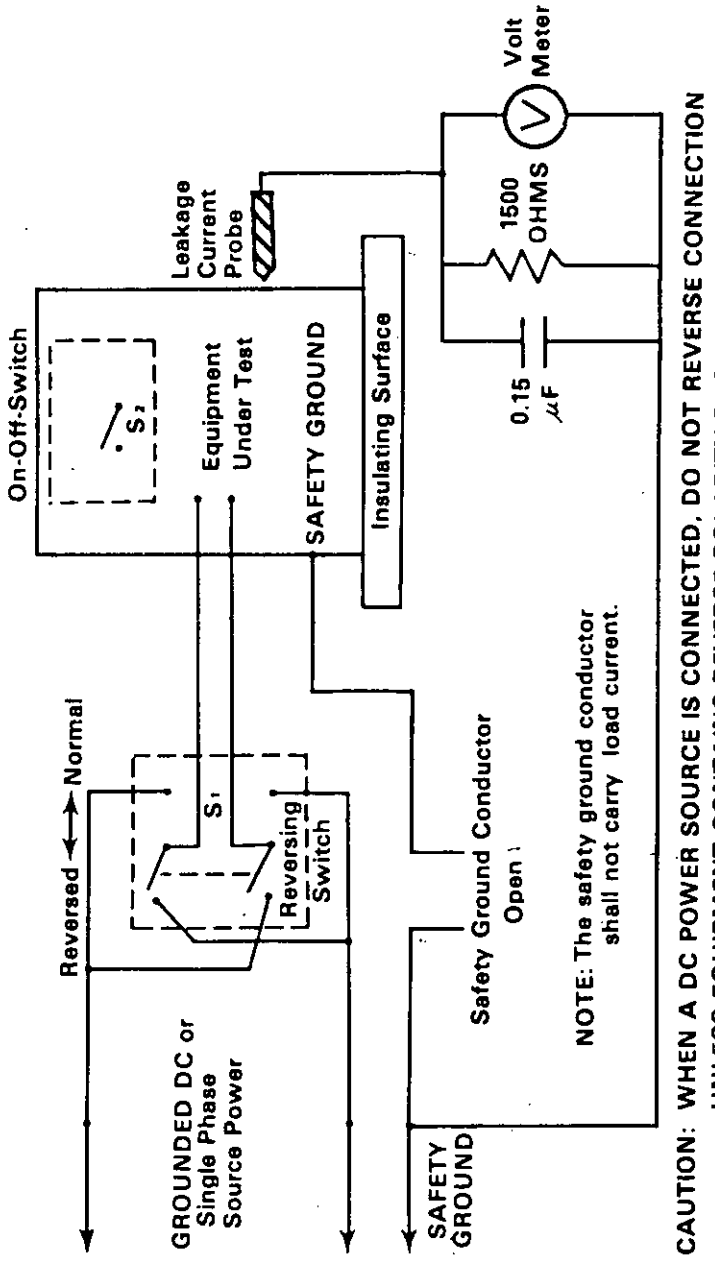
4.9.2 Equipment conditioning. Each deliverable production equipment shall be subjected to operational burn-in for a period of 72 hours under conditions of -28°C to +65°C temperature cycling (approximately three hours at -28°C and five hours at +65°C). The receiver/transmitter shall be operating at 50 percent transmit and 50 percent receive with an average transmitted power of at least 100 watts. Equipment conditioning shall be performed prior to conducting the additional production tests specified in TABLE IV.

4.9.2.1 Equipment configuration. Individual equipments (as specified in 3.3) need not be tested in a system configuration provided provision is made to fully exercise all active electrical and electromechanical elements. The antenna and H-189/U handset may be excluded from the test.

4.9.2.2 Accept criteria. Failures occurring prior to the final 24 hours of equipment conditioning shall be repaired and the burn-in resumed. In the event of failure during the final 24 hours, the equipment shall be repaired and the burn-in shall be resumed with a minimum of 24 hours of successful operation prior to acceptance.

4.9.2.3 Rework. In the event an equipment is reworked following initial successful completion of the burn-in test, and the rework is of a minor nature (such as replacement of a single component or group of components), equipment conditioning need not be reperformed. However, should the rework be major (that is, replacement of a plug-in module with a module which has not previously been burned-in), equipment conditioning shall be reperformed as specified herein.

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FIGURE 1. Single phase or DC test diagram for leakage current measurements.

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4.9.3 Electromagnetic compatibility test. The requirements of 3.4.5 shall be verified by tests performed in accordance with MIL-STD-462. For environmental inspection, only tests CE03 and RE02 shall be performed.

4.10 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1. Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The AN/URC-94(V)-1 is intended for service craft and boats applications. The AN/URC-94(V)-2 is intended for fixed shore installation applications.

6.2. Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples required if other than specified in 4.3
c. Nomenclature of equipment desired
d. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.5.5, 4.6.8	Report failure summary and analysis	DI-R-7041, 21 Oct 1977
4.5.6, 4.6.8	Plan, corrective action	DI-R-7038, 21 Oct 1977
4.3, 4.4.1, 4.4.2, 4.4.3, 4.4.4, 4.5, 4.6, 4.8, 4.9	Test procedures	UDI-T-22710B, 1 Jul 1976

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104, 55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

Preparing activity:
NAVY-EC
(Project 5820-N719)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-R-28848

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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